

Aditya

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EDUCATION

CMR Institute of Technology

B.E. in Artificial Intelligence and Machine Learning

Bangalore, India
Dec 2022 - July 2026

Kendriya Vidyalaya NAL Campus

Senior Secondary

Bangalore, India
June 2020 - June 2022

EXPERIENCE

Zetheta Algorithms Private Limited

Data Science Intern

Oct 2025 - Dec 2025

Remote

- Developed a cross-domain AI pipeline integrating computational material science datasets (AFLOW, Materials Project, NIST) with commodity and infrastructure financial data to model material-driven financial risk and asset valuation.
- Implemented advanced ML architectures including Crystal Graph Neural Networks (CGNNs) for crystal structure representation and Physics-Informed Neural Networks (PINNs) to enforce physical constraints in material property prediction.
- Built a deep survival analysis model (DeepSurv) and engineered cross-domain features linking material properties (fatigue life, corrosion rate, tensile strength) with financial indicators to estimate infrastructure remaining useful life (RUL), credit risk metrics, and commodity risk signals.

Learner's Byte

AI Intern

Jan 2026 - April 2026

Remote

- Developed Python-based GenAI applications using LLM APIs and modern AI frameworks.
- Built Retrieval-Augmented Generation (RAG) systems using vector embeddings and semantic search.
- Integrated external APIs and structured output pipelines for production-ready AI workflows.
- Debugged and optimized AI applications through prompt tuning and response validation.

PROJECTS

Automated Machine Learning Pipeline | Python, Streamlit, Scikit-learn, Pandas, NumPy

- Engineered an end-to-end ML pipeline supporting classification, regression, and clustering using Scikit-learn.
- Automated data preprocessing, feature engineering, and model selection using cross-validation techniques.
- Enabled dynamic dataset input and improved model selection efficiency through an interactive Streamlit interface.

AI-HRMS Platform | FastAPI, PostgreSQL, JWT, Python

- Developed a backend HRMS system using FastAPI and PostgreSQL following RESTful architecture.
- Implemented JWT-based authentication and role-based access control.
- Designed scalable CRUD operations and API documentation using Swagger/OpenAPI.

SKILLS

Languages: Java, Python, SQL, JavaScript, C

Backend Development: FastAPI, REST APIs, Authentication (JWT), DB Design, CRUD Operations

Databases: PostgreSQL, SQL

GenAI and AI: LLMs, RAG, Prompt Engineering, Vector Embeddings, Scikit-learn, TensorFlow, PyTorch

Tools: Git, GitHub, Linux, VS Code, Postman, Jupyter Notebook

Cloud: AWS (EC2, S3), GCP

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Software Engineering, Computer Networks

ACHIEVEMENTS

- Winner of AI-Thon 2025 conducted by BITS-Pilani
- Odoo x Adani University Hackathon '26
- Visa 24hrs AI Hackathon(Indian Institute of Technology, Madras)